TATA TECHNOLOGIES LTD.

TECH PULSE FY-26 | AI/ML Capstone Project

TECHNICAL REPORT

Field	Details
Institute / University	
Programme / Batch	
Student Name	
PRN	
Case Study ID (CS1–CS5)	
Project Title	

1. Problem Statement and Intended Users

2. Input Data / Knowledge Base

Item	Description
Sources	
Size and format	
Preprocessing / EDA done	
Chunking strategy	

3. Solution Architecture

Component	Choice (name, version, hosting)
Pre-trained model	
Embedding model	
Vector store (FAISS / ChromaDB)	
RAG pipeline / AI-agent design	
Service layer and UI	
Tools / deterministic rules	

[Insert system architecture diagram here]

4. Implementation and Configuration

Explain the workflow step by step (ingestion → retrieval → generation → response), with key parameters such as chunk size, top-k, temperature and prompt settings.

[Write here]

[Insert workflow diagram here]

5. Evaluation Steps and Results

Describe the test question set, ground truth and metrics. Report actual results, including failures.

#	Test question / scenario	Expected answer / behaviour	Result (Pass/Fail, notes)
1			
2			
3			
4			
5			

Metric	Score / observation
e.g. Answer relevance / faithfulness	
e.g. Retrieval hit rate / citation accuracy	
e.g. Latency	

6. Responsible AI Measures

Area	Measure adopted in this project
Grounding and citations	
Privacy	

Area	Measure adopted in this project
Security	
Human oversight / review	

7. Innovation Highlights

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8. Screenshots, Logs and Evidence

[Figure 1: caption]

[Figure 2: caption]

9. Reflection and Conclusions

[Write here]

10. AI-Tool Usage Declaration

AI tool / model	Purpose	Artifact affected	Verified By Student (Y/N)

11. Student Declaration

I confirm that this report represents my own understanding and contribution, that the work follows the approved case-study scope, and that I can explain every significant part of the implementation.

Signature:		**Date:**	
Name:		**PRN:**	